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Constraining Reionization: Correlating Ly α Emission with Galaxy Properties

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Executive Summary: By the end of the funding from this grant my work will significantly transform the way we constrain the timeline of reionization. **Project 1:** Using the Texas Euclid Search for Lyman-Alpha (TESLA) survey data I have on hand **now** I will develop a predictive algorithm using thousands of galaxies between redshifts 2 – 3.5 to predict emergent Ly α emission based on SED-derived galaxy properties. **Project 2:** Performing nebular line modeling of *in-hand JWST* NIRSpec spectroscopic data of redshift 2 – 3.5 galaxies to uncover the physical properties within galaxies that regulate the escape of Ly α , providing a complete picture of how galaxies regulate Ly α emission. **Project 3:** Using an expanded TESLA data set and early *Euclid* imaging data to further constrain and robustly train my predictive algorithm from **Project 1**. **Project 4:** Directly compute the neutral hydrogen fraction using the algorithm from **Project 3** on galaxies between redshifts 6 – 10 using publicly available (and proposed-for) NIRSpec data and infer the timeline of reionization.

Introduction: Understanding the timeline of reionization can tell us about the earliest ionizing sources in the universe, and allow the study of large-scale structure formation and evolution. The timeline of reionization can be inferred by measuring the neutral hydrogen fraction during the epoch of reionization (EoR). Figure 1 shows the predictions of the neutral fraction between two competing reionization models, where reionization is driven by bright or faint galaxies. By measuring the neutral hydrogen fraction during a period where these two models are the most different we can infer the timeline for reionization. Ly α is the most promising tracer we have at the moment as it is observed in star-forming galaxies and these galaxies are abundant in the early universe (e.g., [1]; [6]), implying that Ly α may be able to trace the spatial nature of reionization. It is a sensitive tracer, as Ly α photons are resonantly scattered, decreasing their observability when they have to traverse through neutral gas. Thus, I will use the Lyman-alpha (Ly α) emission line to measure the neutral fraction above redshift 6 to better constrain the timeline of reionization.

However, in order to use Ly α to trace reionization we need to know **how much** Ly α flux emerges from a galaxy to compare against the observed flux, as any differences are attributed to neutral hydrogen randomly scattering the Ly α photons. To date, **no good metric for acquiring the emergent Ly α flux of galaxies exists**. Getting a robust estimate of the emergent Ly α flux is complicated by processes within a galaxy, such as dust content, gas kinematics, and dust distribution which affect the Ly α strength that ultimately escapes from a galaxy (e.g., [12], [10], [3] and references therein). Previous studies [e.g. 16, 11, 8] assume that the Ly α equivalent width (EW) distribution at $z \approx 6$ is able to predict the emergent Ly α flux of a galaxy above redshift 6. This is not a robust assumption as studies of galaxies at redshift 6 are limited by small sample sizes (typically ~ 100 -1000 sources). Spectroscopic follow-up usually requires a pre-selection of sources, which biases studies toward the most luminous sources. There are also significant differences in selection criteria between studies so every study analyzes different galaxy populations. To overcome these challenges I will create a **larger, unbiased-selected** sample of galaxies to tie back the emergent Ly α emission to the physical properties of an individual galaxy. I will use

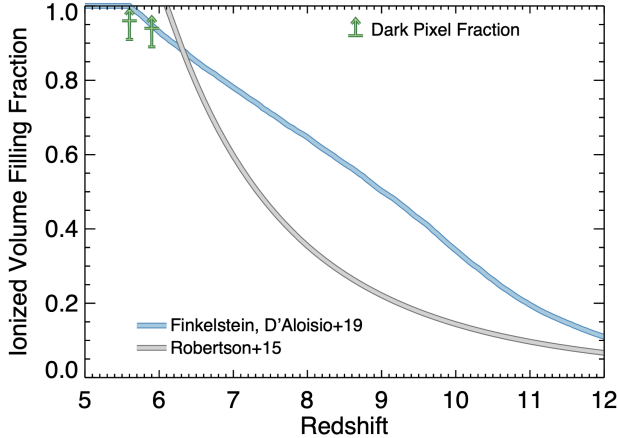


Figure 1: Figure from [5] that shows the ionized fraction as a function of redshift between competing models. The model where reionization is driven by small faint galaxies is shown in blue, while a model where bright massive galaxies drive reionization is in gray. These two models predict drastically different neutral hydrogen between redshifts 7 – 10. By measuring the neutral fraction at these redshifts via Ly α emission I will be able to infer which ionizing sources drove reionization and constrain the timeline of reionization.

the results of this analysis to train a predictive regression algorithm **to better predict a galaxy’s emergent Ly α emission based on the observed properties of individual galaxies** and use this algorithm to compute the neutral fraction at $z = 6 - 10$ to **better constrain the timeline of reionization**.

Leveraging the TESLA Survey to study Ly α as a Probe of Reionization

The TESLA survey will significantly improve our ability to constrain the timeline of reionization by leveraging thousands of Ly α -emitting galaxies with a wide range of galaxy properties. TESLA is a **large, unbiased**, spectroscopic survey *I am leading* that is observing the 10 deg^2 *Euclid* North Ecliptic Pole (NEP) field. TESLA uses the Visible Integral-Field Replicable Unit Spectrograph (VIRUS) [7] on the Hobby Eberly Telescope. VIRUS covers 3500 – 5500 Å with 78 IFUs spanning $\sim 18' \times 18'$, allowing for optimal Ly α detection in galaxies between redshifts 1.9 – 3.5, where IGM attenuation of Ly α is minimal. The status of the TESLA survey is shown in Figure 2, where we have $\sim 40\%$ of the field covered **now**, with the survey expected to be fully completed by Fall 2025 (the start of the third year of my fellowship). The NEP field has a wealth of deep multi-wavelength imaging already available from the Hawaii 20 degree (H20) survey and soon will have *Euclid* near-infrared (NIR) imaging and photometry. The H20 data includes Subaru Hyper Suprime-Cam *grizy* and *Spitzer* IRAC imaging and photometry. The H20 survey is $> 90\%$ complete, with the remaining needed nights expected to be allocated prior to the first year of funding (Spring/Summer 2023). Through collaboration with the H20 team, I have access to fully reduced imaging and photometric catalogs which are pivotal in acquiring a galaxy’s global properties through SED fitting and using my emission line classification methodology [14]. This combination of H20 + TESLA allows for accurate measurements of the emergent Ly α flux from TESLA galaxies.

To properly train an algorithm to predict the emergent Ly α strength of galaxies I require a large sample of galaxies with a wide range of galaxy properties to sample a broad dynamic range in galaxy stellar population properties. While TESLA will ultimately provide 1000’s of Ly α -selected galaxies, my pilot study (with 43 such galaxies) has shown that galaxies in the TESLA survey span a large dynamic range in various galaxy properties, (e.g. stellar mass, dust, SFR, and more) and that correlations do exist between galaxy properties and Ly α strength [14]. The results imply that with a larger sample of galaxies, I can leverage correlations between galaxy properties and Ly α EW to predict their emergent Ly α strength.

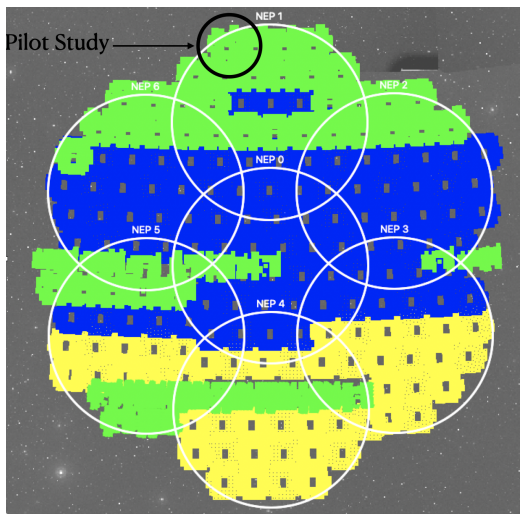


Figure 2: The 10 deg^2 North Ecliptic Pole field. My pilot study focuses on the black circle at the top of the image and consists of 2 HETDEX pointings with each pointing consisting of an $18' \times 18'$ rectangular region being observed with ~ 300 IFUs (4×78 IFU sub-pointings). The in-hand HETDEX data is outlined by the green regions, while the white circles are the Hyper Suprime Cam-H20 imaging survey which includes deep optical imaging in the g, r, i, z, y bands. We expect that the NEP field will be $> 100\%$ covered by VIRUS (denoted by the green+blue regions) by the start of the third year of funding.

Testing Feasibility in TESLA: This past year I conducted a pilot study on a portion of the NEP field where the H20 and TESLA surveys had overlapping data, shown in the upper left of Figure 2, to test the feasibility of using correlations between galaxy properties and Ly α EW into a predictive algorithm [14]. I developed a methodology to find galaxies emitting Ly α emission and fit their stellar populations using the Bayesian SED-fitting code **BAGPIPES** to get their galaxy properties[2]. Using my methodology I discovered 43 galaxies and uncovered that the most correlated galaxy properties with Ly α EW are stellar mass and star formation rate (SFR) with a Spearman correlation coefficient of $-0.34^{+0.17}_{-0.14}$ and $-0.37^{+0.16}_{-0.14}$, indicating that these two properties can be used to predict the emergent Ly α emission of galaxies. However, a much larger sample is needed to increase the significance of the correlations, which I will do in my first year of the fellowship by significantly expanding the analysis to include more of the green VIRUS data in Figure 2, substantially increasing the sample size to >1000 Ly α -detected galaxies. The increased sample size will provide more robust correlations between galaxy properties and Ly α EW and will be used to train an algorithm to predict the emergent Ly α emission. This will result in **one published paper** that will outline the expanded sample size, methodology, and a study of their galaxy properties, and introduce the algorithm to predict emergent Ly α strength.

***Euclid*:** *Euclid*, launching in the summer of 2023, will observe the NEP field deeply over its mission. My PI is one of the few U.S. scientists to have access to proprietary *Euclid* near-infrared (NIR) YJH imaging that can be used in my TESLA analysis. The YJH filters are crucial in constraining galaxy properties as they probe the Balmer break of galaxies between redshifts 2 – 3.5, breaking the dust-age degeneracy. Prior studies have found there to be an anti-correlation between Ly α EW and dust (e.g. [19, 17, 20]), which the inclusion of *Euclid* data can improve my derived correlations between Ly α EW and dust. By the time *Euclid* completes its first year of observations in NEP, TESLA will be nearly complete – later in my fellowship I will use this increased TESLA data and *Euclid* photometry to **publish** a paper updating my predictive Ly α algorithm inclusive of the *Euclid* data and the expanded sample size. Since NASA is collaborating with *Euclid*, the NASA-funded project will play a pivotal role in better constraining the timeline of reionization by better constraining the

SED-derived properties of galaxies and more accurately calibrate the algorithm to predicting the emergent Ly α strength based on galaxy properties.

Training Algorithm Above Redshift 3.5: The analysis of TESLA will train an algorithm that is well suited to predict the emergent Ly α emission of galaxies between redshifts 2 – 3.5. However, this algorithm needs to work above redshift 6 to measure the neutral fraction during EoR. To accurately predict emergent Ly α flux from higher redshifts galaxies I will leverage existing higher redshift studies of galaxies with detectable Ly α emission to train the algorithm against (e.g. [8, 18, 4]), adding in redshift-dependent terms as needed. This is crucial in making robust measurements of the neutral fraction above redshift 6 to distinguish between competing reionization timelines. **I will use this algorithm in my 4th Project which will measure the neutral fraction between redshifts 6 – 10 (see below).**

NEP Summary: This portion of my proposed work will result in two published papers: one outlining the correlations between galaxy properties and Ly α strength using thousands of galaxies between redshifts 2 – 3.5 and introducing my algorithm to predict the emergent Ly α flux based on galaxy properties, expected early 2024. The second paper will expand upon the first paper by combining the full TESLA field including early *Euclid* YJH photometry to improve emission line identification, better constrain the SED-derived galaxy properties, and improve my predictive algorithm, expected by late 2025.

Nebular Line Modeling of $1.9 < z < 3.5$ Galaxies using NIRSpec

Galactic spectra allow for detailed measurements of the underlying stellar population and the driving physical properties leading to observed emission and absorption lines. As a member of the Cosmic Evolution Early Release Science (CEERS) team, I have access to fully reduced *JWST* NIRSpec data **now** from CEERS Cycle 1 data. I will use NIRSpec spectra to perform nebular line modeling of galaxies in the Extended Groth Strip (EGS) field between redshifts 1.9 – 3.5 to measure the internal properties of galaxies. To study how internal galaxy properties promote Ly α escape I will use the HETDEX survey, as HETDEX is probing the Ly α emission of galaxies at these redshifts. There are ~ 15 galaxies that have both HETDEX Ly α detections and CEERS NIRSpec spectra. To increase the sample, I will use any relevant guaranteed time observation data once it becomes available in 2023, as the NIRSpec team’s Wide survey will also be observing several HETDEX-observed CANDELS fields. Once all the data is collected, I expect to have spectra for over 50 galaxies to perform this analysis on, and this will result in **one published paper studying the internal properties of galaxies that promote the escape of Ly α emission.**

This project will focus on exploring the internal properties of galaxies to answer *why* Ly α is emitting from these galaxies. The NIRSpec instrument operates between 1 – 5 microns (μm), probing the rest-frame optical of galaxies between redshifts 1.9 – 3.5. The rest-frame optical spectra of galaxies have many diagnostic emission lines that can be used to infer the physical properties of a galaxy. Figure 3 is a $z \approx 2$ NIRSpec spectrum of a galaxy and shows prominent diagnostic emission lines that we can use to perform nebular line modeling. Instead of using empirical line ratio analysis, which has assumptions in their calculations or is correlated with other parameters [15], I will be using line ratios determined by MAPPINGS [9] which incorporates temperature gradients, a realistic representation of atoms, and a self-consistent theoretical model that incorporates the coupling of many parameters. I built an MCMC algorithm to robustly sample all the MAPPINGS parameters to determine the ISM

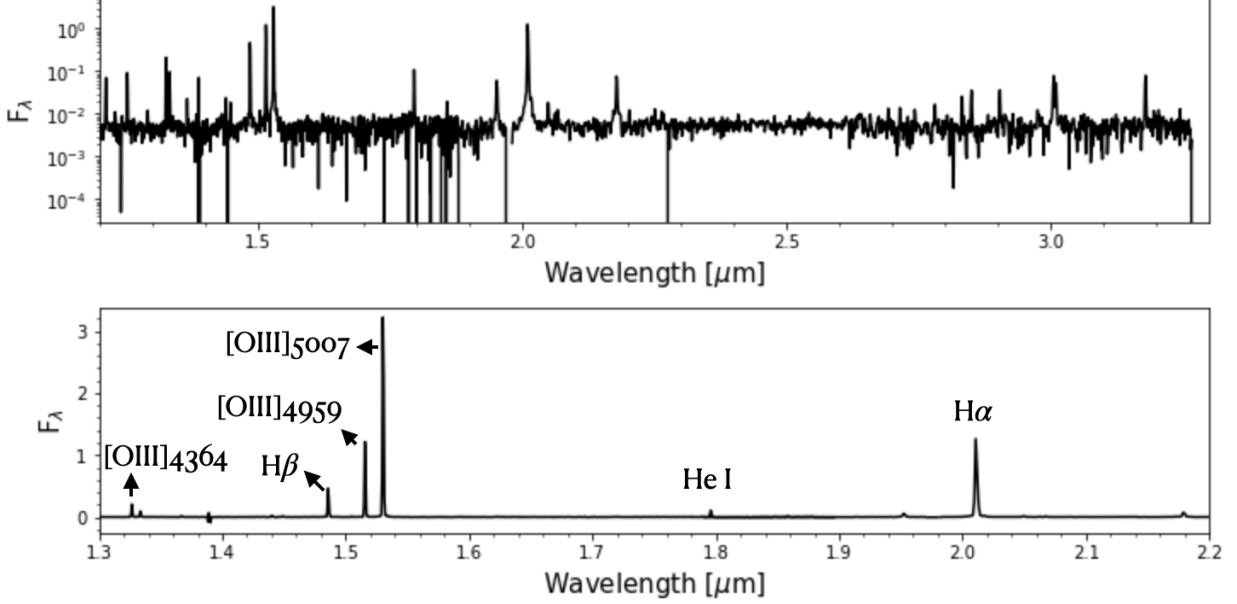


Figure 3: CEERS JWST NIRSpec spectra from a $z \approx 2.03$ HETDEX LAE. We can see the wealth of lines such as H β , [OIII]5007, 4959, 4363, and even probing H α emission which is useful for emission line diagnostic measurements.

properties that best reproduce the ratios of each galaxy. This is pivotal in understanding which property or set of properties aids or hinders the escape of Ly α within a galaxy. **This analysis will result in a paper explaining what properties promote or hinder the escape of Ly α , and when used with Project 1 and 3 will tell the complete story of how Ly α is able to escape from galaxies.**

Measuring the Neutral Hydrogen Fraction During Reionization

With *JWST* NIRSpec we are now able to detect Ly α emission from galaxies during EoR [18]. Ly α can be used **now** to measure a wide range of neutral fractions throughout reionization. However, the main limitation in this analysis is that no good metric for acquiring the emergent Ly α flux of galaxies exists. This will change when I complete **Project 1 and 3**, expected in late 2025, as I will have a robust algorithm to predict the emergent Ly α emission based on a galaxy's physical properties.

I will then use direct observations of $z > 6$ galaxies to apply my algorithm directly to measure neutral fractions. I will use relevant JWST NIRCам and MIRI imaging and photometry to select high redshift galaxies and use any relevant photometric catalogs to acquire their global galaxy properties through SED-fitting. To get the Ly α emission from these sources I will cross-match high redshift candidates to any relevant public NIRSpec observations, as Cycle 1 had many NIRSpec observations taken of high redshift galaxies (and Cycle 2 likely will as well). I will supplement these data sets by **leading a *JWST* Cycle 3 proposal to acquire Ly α emission**. Having a substantive high-redshift galaxy population is crucial to average out line-of-sight variations of an inhomogeneous reionization to get a sense of the average Ly α IGM transmission, and hence the neutral hydrogen fraction, at various redshift intervals. I will compare these measurements against the model predictions from Figure 1 to infer the timeline of reionization. This analysis will result in **one published paper outlining the computation of the neutral fraction as a function of redshift**

and my analysis to better contain the timeline of reionization.

We note that the PI has a current ADAP proposal to fund research in the NEP field but the FINESST proposal goes beyond what NASA has agreed to support in several ways: development of a novel technique based on SED modeling to classify detected emission lines, the inclusion of a theoretical comparison component, of early *Euclid* data to improve both classification and analysis, and the inclusion of my *JWST* spectroscopic project.

Relevance to Astrophysics Research Program

These projects are directly related to several stated objectives of NASA’s science mission directive (SMD) 2014 Science Plan (e.g. see Chapter 4) as well as those listed in the 2020-2024 A Vision for Scientific Excellence strategy 1.1. The physics of the cosmos: (1) Discover how the universe works (2) Explore the origin and evolution of galaxies, stars, and planets that make up our universe, Cosmic Origins Program: (3) Understanding how the universe evolved since the Big Bang. The grant will support research that addresses the timeline of reionization which directly ties in with the evolution of the universe objective outlined above. By studying the physical properties of galaxies between $z \sim 1.9$ -3.5 we will explore any redshift-dependent properties that follow the galactic evolution objective. Once my predictive model is finished, we will be able to better understand the role galaxies played during reionization and answer how the universe evolved during this epoch. Using NASA-funded instruments such as the *JWST* to perform nebular line modeling to determine the exact physical mechanism(s) that regulate the escape of $\text{Ly}\alpha$ in galaxies at $z \sim 3$ and *Euclid* NIR photometry to break the age-dust degeneracy will demonstrate the affect the NIR has on my predictive model and improve the algorithm. Both of these NASA-funded instruments will help me gather a complete picture of how $\text{Ly}\alpha$ escapes from $\text{Ly}\alpha$ -emitting galaxies.

Timeline: From the time of funding from this grant I plan on publishing **4** papers.

Year 0: Spring 2023 - Submitted the NEP Pilot study to ApJ. I will work on scaling my code to handle the current TESLA data on hand now, green pointings in Figure 2. Concurrently, I will be getting code up and running to analyze NIRSpec data of galaxies between redshifts 2 – 3.5 in existing CEERS data.

Year 1 - Start and **publish** a paper on the first NEP project using an increased sample size of > 1000 galaxies and introduce the predictive algorithm. I will analyze roughly 50 HETDEX-selected galaxies that have NIRSpec spectra and work on implementing my MCMC algorithm to sample the various galaxy parameters. I will compare the galaxy properties to the $\text{Ly}\alpha$ emission from HETDEX to explore any correlations. At this stage, I will write the *JWST* Cycle 3 proposal to search for more galaxies with $\text{Ly}\alpha$ detections between redshifts 6 – 10.

Year 2 - Publish a paper on nebular line modeling to infer galaxy properties and correlation to $\text{Ly}\alpha$ emission in the NIRSpec and HETDEX data. I will then work on implementing *Euclid* NIR photometry to constrain the physical properties of galaxies and robustly train my algorithm to predict $\text{Ly}\alpha$ flux based on galaxy properties based on the full TESLA sample. Concurrently I will be training the algorithm using higher redshift studies to be used to measure the hydrogen fraction during reionization.

Year 3 - I will finish up any final analysis in the final NEP project and **publish** the results and predictive algorithm including *Euclid* data. I will compute the neutral hydrogen fraction during reionization and **publish** a paper on the results of this analysis, and start working on my dissertation and defense.

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Acknowledgement

The (FI) Oscar A. Chavez Ortiz authored this proposal, including the Personal Statement and the Research Statement. I received editorial support from the Principal Investigator (PI) Steven Finkelstein, my friends and colleagues Gene Leung, and Rebecca Larson, in order to improve grammar, clarity and overall structure of the proposal!!

Research Readiness Statement

Acquiring the skills necessary in order to correlate galaxy properties to Ly α emission strength has been a process I have been working on throughout my academic career. During my undergraduate studies, I made sure to take relevant courses in extra-galactic research by taking a graduate-level galaxies course to enhance my knowledge of galaxies and galaxy evolution. I also worked with Mariska Kriek on a project related to stellar synthesis modeling of galaxies in order to determine their global properties. That research experience gave me valuable scientific skills such as error propagation, spectral continuum removal, identification of emission lines, and line fitting techniques.

In the Summer of 2019 I did research to identify whether 12 low redshift galaxies ($z < 0.5$) were viable candidates for being analogs of galaxies during the epoch of reionization. The task involved making my own pipeline that reduced the 2-dimensional spectral data into a 1-dimensional spectrum, continuum removal to acquire accurate flux measurements for line ratio diagnostics. I was able to calculate the equivalent width (EW) of these sources and compare them with known Lyman-continuum leakers from other low redshift studies. I found that these galaxies were great analogs of high-redshift galaxies, exhibiting properties very similar to known Lyman-continuum leakers. This work resulted in an American Astronomical Society Research Note [13]. The key takeaway from this project was using emission lines to derive galaxy properties of interest, which is going to help me with my 2nd paper using JWST NIRSpec data when performing my emission line modeling analysis.

My time during graduate school has been actively used to understand my data and utilize the resources at my disposal to help me progress in my research projects. My first year was spent learning what an LAE looked like in the TESLA dataset and learning the characteristics that separate them from [OII]-emitters. I also got familiar with the SED fitting code, BAGPIPES, by taking test galaxies and fitting them to acquire global galaxy properties. I learned how to utilize the resources at the Texas Advanced Computing Center (TACC) to help me achieve my research goals, such as accessing the TESLA data and running jobs on their supercomputer. At the same time, I made sure to take courses relevant to my research by taking a galaxies course, and a cosmology and large-scale structure course.

My second year in graduate school was spent automating my data process pipeline and scaling my code to handle a sample size that would be 10x larger. As of the writing of this proposal, I have in place a pipeline that is able to take a sample of VIRUS spectra, match them to an imaging counterpart, determine if the source is an LAE, and fit them with BAGPIPES to get galaxy properties. I also took the time to take courses designed to enhance my research project by taking an astronomical data science course, learn statistics, and Bayesian techniques. I also took radiative transport to better understand the mechanisms involved that enhance and/or hinder the escape of Lyman-alpha in a galaxy.

During my third year in graduate school, I have gotten familiar with the output from the MAPPINGS nebular line fitting code. I have designed an algorithm to use the results of MAPPINGS and run them using Markov Chain Monte Carlo (MCMC) to acquire the best set of properties for a given set of line ratios. I also worked with JWST NIRSpec data and used my MCMC MAPPINGS algorithm to find the set of parameters for every galaxy I have and see if there are any correlations that promote or hinder Lyman-alpha escape.

I believe I am capable of carrying out the research I have stated in the main proposal and on track to finish my Ph.D. in three years, by the end of the funding of this grant (08/2026).

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RESEARCH INTERESTS

My research interest revolves around trying to understand how we can correlate galaxy properties to Lyman-alpha emission so that we can use Lyman-alpha as a probe of reionization. I am leading the analysis of the TESLA survey where we are attempting to gather $\approx 50,000$ Lyman-alpha emitting galaxies (LAEs) so that we can explore correlations between galaxy properties and emerged Lyman-alpha strength between redshifts 2 – 3.5. We will expand this analysis to higher redshift by incorporating higher redshift LAE studies to make a predictive algorithm of emerged Lyman-alpha strength based on SED-derived galaxy properties. I also am part of the CEERS team and will use NIRspec to study the internal mechanisms of LAEs to understand **why** Lyman-alpha is able to escape from these galaxies. I will finally be able to put my predictive algorithm to use by studying galaxies at the epoch of reionization era and computing the neutral hydrogen fraction beyond redshift 6.

EDUCATION

Ph.D. in Astronomy

University of Texas at Austin (UTA)
Department of Astronomy
2515 Speedway, Stop C1400
Austin, Texas 78712-1205
Advisor: Dr. Steven Finkelstein

Expected Spring/Summer 2025

B.A. in Astrophysics

The University of California Berkeley (UCB)
501 Campbell Hall 3411
Berkeley, CA 94720-3411
Advisor: Dr. Mariska Kriek

Fall 2019

PROJECTS

Project 1 NEP Pilot Study: I was tasked with looking at a small patch of the North Ecliptic Pole field and testing the feasibility of a much larger analysis. My work involved using spectroscopic and photometric data to come up with a set of criteria to find a specific type of galaxy called Lyman-alpha Emitters and then study how their global galaxy properties correlate with emerged Lyman-alpha strength. For this project, I had to come up with a scalable code that would work on the small survey area I was working with but also work on an expanded data set once more data became available in an automated way. This work is in the process of being published in the Astronomical Physical Journal, a peer-reviewed article, for anyone to look into the analysis.

Project 2 Spectroscopic Analysis of LAEs: For this project I was tasked with analyzing spectroscopic data from a publically available data set to complement my Project 1 work. By looking at the spectroscopic information of galaxies we are able to study the internal properties of these galaxies. Having an in-depth understanding of the internal properties of galaxies is crucial in understanding which properties help or hinder the escape of certain emission lines. For our purposes, we were trying to study which properties promote and/or hinder the escape of Lyman-alpha. This

project had me design an MCMC algorithm with the package `emcee` to sample various parameters to come up with the parameters that best matched our observational data.

Project 3 Spectroscopic Analysis of Nearby Galaxies: During the summer of 2019 I was tasked with spectroscopically analyzing 12 nearby galaxies. The goal of this project was to determine if these 12 galaxies could be used as analogs for very distant galaxies. For this project, I had to reduce the raw spectroscopic data and come up with a 1-dimensional spectrum for me to then analyze any emission lines. In this project, I learned how to perform model fitting as I had to fit the emission lines with a Gaussian profile to get the flux for each galaxy. I did this using Monte Carlo, by perturbing the flux and creating 1000 realizations of the spectrum and fitting the lines, and using the resulting flux distribution to get our average flux and quantify our errors. At the end of this analysis, we were able to conclude that these 12 galaxies are very likely candidates of very distant galaxies and that a follow-up of these sources was needed.

HONORS & AWARDS

Mentoring Dean's Strategic Fellowship	2020 – 2022
Daniel Edward Wark Memorial Scholarship	Summer 2018
Dean's Honor List at Riverside City College	2016 – 2017

AWARDS & GRANTS

FY20-22	Mentoring Dean's Strategic Fellowship	\$68K
FY18-19	Daniel Edward Wark Memorial Scholarship	\$10K

CONFERENCES & PRESENTATIONS

Science Presentations

Poster: American Astronomical Society #241 (SEATTLE, WA)	12 January 2023
Talk: Euclid Consortium	18 September 2022
Poster: American Astronomical Society #240 (PASADENA, CA)	14 June 2022
Talk: STSCI MOS	May 2022
Invited Talk: Yale Lunch Talk	28 April 2022
Poster: American Astronomical Society #235 (HONOLULU, HI)	5 January 2020

TEACHING EXPERIENCE

Teaching

— Instructor, Python for Astronomers DeCal	UC Berkeley, Fall 2019
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Workshops

— VIP Python BootCamp Instructor	January 2023 — UT Austin
— REU/TAURUS Python BootCamp, Instructor	June 2022 – UT Austin
— Co-organizer of Python BootCamp	June 2021 – UT Austin

Assistant

— Teaching Assistant, Astronomy	UT Austin, Spring 2021
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PROGRAMMING

Languages

Fluent: Python, LaTeX, Astropy, Scipy, Numpy, EAZY, BAGPIPES

Experience with: C++, bash, R, HTML, SQL, Machine Learning

Steven L. Finkelstein

CONTACT INFORMATION	The University of Texas at Austin Department of Astronomy 2515 Speedway, Stop C1400 Austin, TX 78712	Office: (512) 471-1483 stevenf@astro.as.utexas.edu www.as.utexas.edu/~stevenf
RESEARCH INTERESTS	I study how galaxies evolve from small galaxies in the very the distant universe, to the large, morphologically complex Milky Way today. I focus on finding the most distant galaxies in the universe, confirming their distances, characterizing their nature, and investigating their impact on the universe around them. This requires a combination of deep imaging from the <i>Hubble Space Telescope</i> with followup spectroscopy from the largest ground-based telescopes.	
ACADEMIC POSITIONS	The University of Texas at Austin , Austin, TX Professor Fall 2022 – Present Associate Professor Fall 2017 – Summer 2022 Assistant Professor Fall 2012 – Summer 2017 The University of Texas at Austin , Austin, TX Hubble Fellow Sept. 2011 – Aug. 2012 • Faculty Contact: Professor Karl Gebhardt Texas A&M University , College Station, Texas Sept. 2008 – Aug. 2011 Postdoctoral Research Associate • Faculty Advisor: Professor Casey Papovich	
EDUCATION	Arizona State University , Tempe, Arizona Ph.D. Physics, Emphasis in Astronomy August 2008 • Advisor: Professor James E. Rhoads • Dissertation: Physical Properties and Dust Effects in High-Redshift Lyman Alpha Galaxies University of Washington , Seattle, Washington B.S. Astronomy and Physics June 2003	
PUBLICATIONS	Summary: >220 papers published in or submitted to peer-reviewed journals, with an h-index of 61. Of these, I am the lead author on 23 papers, which have ~2600 citations combined. Some highlights are: 1. Finkelstein, S. L. et al. 2023, <i>CEERS Key Paper I: An Early Look into the First 500 Myr of Galaxy Formation with JWST</i>, Astrophysical Journal Letters, in press 2. Finkelstein, S. L. et al. 2022, <i>A Long Time Ago in a Galaxy Far, Far Away: A Candidate $z \sim 12$ Galaxy in Early JWST CEERS Imaging</i>, Astrophysical Journal Letters, 940, 55 3. Finkelstein, S. L. et al. 2021, <i>A Census of the Bright $z = 8.5\text{--}11$ Universe with the Hubble and Spitzer Space Telescopes</i>, Astrophysical Journal, in press 4. Finkelstein, S. L. et al. 2019, <i>Conditions for Reionizing the Universe with A Low Ionizing Photon Escape Fraction</i>, Astrophysical Journal, 879, 36	

5. *Invited Review Article: Finkelstein, S. L. 2016, Observational Searches for Star-Forming Galaxies at $z > 6$* , Publications of the Astronomical Society of Australia, 33, 37
6. Finkelstein, S. L., Ryan R, Papovich, C. et al. **2015**, *The Evolution of the Galaxy Rest-Frame Ultraviolet Luminosity Function over the First Two Billion Years*, Astrophysical Journal, 810, 71

HONORS AND AWARDS

- 2020 UT Austin Provost's Teaching Fellowship
- 2017-2018 Dads' Association Centennial Teaching Fellowship
- 2016 UT Austin College of Natural Sciences Teaching Excellence Award
- 2015-2016 McDonald Observatory Board of Visitors Teaching Excellence Award
- 2011 Hubble Prize Postdoctoral Fellowship

SELECTED PROFESSIONAL EXPERIENCE

Associate Department Chair (2019 – present)
 Principal Investigator of Approved *JWST* Early Release Science Program (CEERS) and Large Cycle 1 program (NGDEEP)
 Member, Hubble Space Telescope Users Committee (2018 – 2021)
 Member, NASA Cosmic Origins Executive Committee (2018 – 2021)
 Referee for ApJ, ApJL, MNRAS, A&A, Science and Nature

RECENT INVITED TALKS

1. CCA Reionization on a Blackboard, NY, NY, September 2022
2. Santa Cruz Galaxy Formation Workshop, August 2022
3. EAS Annual Meeting (remote), June 2022
4. Hebrew University of Jerusalem, Seminar (remote), November 2022
5. Flatiron Institute CCA, Seminar, May 2022
6. UC Riverside, Colloquium, April 2022
7. Colby College, Colloquium, Sept 2021

GRADUATE STUDENT ADVISING EXPERIENCE

- 2021 – Present: Advisor of graduate student Alexa Morales, who is studying the evolution of galaxy UV colors from $z = 2-14$.
- 2020 – Present: Advisor of graduate student Oscar Chavez Ortiz, who is studying the physical properties of high-redshift galaxies identified by HETDEX.
- 2020 – Present: Advisor of graduate student Katie Chworowsky, who is studying massive quenched galaxies at high redshift.
- 2016 – Present: Advisor of current graduate student Rebecca Larson, who is using *HST* and ground-based spectroscopy to look for Ly α in the epoch of reionization.
- 2013 – 2019: Advisor of Intae Jung, who studied reionization via Ly α spectroscopy.
- 2013 – 2019: Advisor of student Matthew Stevans, who is studying the growth of galaxies using a 24 deg² K-band imaging survey.
- 2013 – 2018: Co-advisor of Jason Jaacks, who is using simulations to make predictions for a JWST Deep Field.
- 2012 – 2016: Advisor of Mimi Song, who focused on the stellar mass functions of very high-redshift galaxies.

Current and Pending Statements

F.I. Oscar A. Chavez Ortiz

1. *Pending*: NASA FINESST

- Asking for \$50,000/year, (\$40,000 salary and \$10,000 for tuition expenses) for three years of funding to support me in my projects to uncover the galaxy properties that hinder and promote Ly α emission.
- Person-months for FI: 12 months/year from 2023-2026
- Start/End Dates: 09/01/2023 - 08/31/2026
- Location: The University of Texas at Austin

Current and Pending Support

P.I. Steven Finkelstein

- *Current:* HST Cycle 27 GO – “Confirmation of a Large, Robust Sample of $z=9-10$ Galaxies in the CANDELS Fields”
 - Awarded for \$63,874 to fund the reduction and analysis of 14 orbits of F098M and F140W imaging of the brightest known $z \sim 9$ candidate galaxies.
 - Person-months for PI: 1.0
 - Start/End Dates: 11/01/2019 – 12/31/2023
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
- *Current:* NSF AAG – “Leveraging the Hobby Eberly Telescope Dark Energy Experiment to Understand Ly α Emission, Galaxy Evolution, and Reionization”
 - Awarded for \$459,079 to UT Austin to support investigations into Ly α emission escape with HETDEX.
 - Person-months for PI: 0.83 summer months per year for three years.
 - Start/End Dates: 9/1/2019 – 8/31/2023
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
- *Current:* NSF AAG – “The Onset of Star-Formation Quenching in Massive Galaxies in the Early Universe”
 - Proposed for \$239,374 to UT Austin to support investigations into quenching in high-redshift galaxies with a deep and wide imaging survey.
 - Person-months for PI: 1.75 summer months over three years.
 - Start/End Dates: 9/1/2020 – 8/31/2023
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
- *Current:* 2021 NASA ADAP – “Improving Lyman-alpha based Constraints on Reionization with Spitzer and HETDEX” (Finkelstein PI)
 - Awarded \$495,417 to support an analysis into the physical properties regulating Ly α photon escape and the properties of massive protoclusters, combining VIRUS spectroscopic surveys with Spitzer/IRAC and ground-based imaging in the NEP and COSMOS fields.
 - Person-months for PI: 2 months over 2021–2023
 - Start/End Dates: 6/1/2022 – 5/31/2025
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
- *Current:* JWST Cycle 1 ERS Post-Launch – “The Cosmic Evolution Early Release Science Program”
 - Awarded \$177,000 to UT Austin to support collection, reduction, analysis and dissemination of data for this approved JWST ERS program.
 - Person-months for PI: 0.75 summer months per year for FY22 and 0.37 for FY23.
 - Start/End Dates: 11/1/2022 – 10/31/2025
 - Location: The University of Texas at Austin

- PI: Steven Finkelstein
- *Current:* JWST Cycle 1 GO – “PRIMER: Public Release IMaging for Extragalactic Research (PI Dunlop)”
 - Awarded for \$59,727 to UT Austin to support simulations of PRIMER NIRCam imaging over the COSMOS field.
 - Person-months for Finkelstein: 0
 - Start/End Dates: 5/1/2022 – 4/30/2025
 - Location: The University of Texas at Austin
 - PI: Jim Dunlop (Local PI Steven Finkelstein)
- *Current:* JWST Cycle 1 GO – “The Next Generation Deep Extragalactic Exploratory Public (NGDEEP) Survey: Feedback in Low-Mass Galaxies from Cosmic Dawn to Dusk”
 - Awarded \$309,297 to UT Austin to support reduction, cataloguing and high-redshift science with NIRCam and NIRISS parallel deep fields.
 - Person-months for PI: 0.75 summer months in FY23 and FY24.
 - Start/End Dates: 2/1/2023 – 1/31/2026
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein (Co-PI’s Nor Pirzkal and Casey Papovich)
- *Current:* JWST Cycle 1 GO – “Confirming a Potential Ultra-Massive Galaxy at $z=10.57$ ”
 - Awarded \$60,304 to UT Austin to support reduction and analysis of a NIRSpec fixed-slit spectroscopic observation of an exceptional galaxy at $z = 10.57$.
 - Person-months for PI: 0.25 summer months in FY23.
 - Start/End Dates: 5/1/2022 – 4/30/2025
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
- *Current:* JWST Cycle 1 GO – “The JWST-legacy narrow-band survey of H-alpha and [OIII] emitters in the epoch of reionization”
 - Awarded \$96,010 to UT Austin to support reduction, cataloguing and analysis NIRCam narrowband and F090W imaging in the CEERS EGS field.
 - Person-months for Finkelstein: 0.3 summer months in FY22 and FY23.
 - Start/End Dates: 5/1/2022 – 4/30/2025
 - Location: The University of Texas at Austin
 - PI: Eduardo Banados (Local PI Steven Finkelstein)
- *Current:* JWST Cycle 1 AR – “Leveraging Early Public JWST Data to Measure Luminosity Functions and Rest-UV Slopes from $6 < z < 12$ ” (**Finkelstein Admin PI**)
 - Awarded \$114,627 to UT Austin to support archival investigations into the UV luminosity function from multiple ERS/GTO datasets. PI is Micaela Bagley, postdoc in Finkelstein’s group.
 - Person-months for Finkelstein: 0.25 summer month total across FY22-FY23.
 - Start/End Dates: 5/1/2022 – 4/30/2025
 - Location: The University of Texas at Austin
 - PI: Micaela Bagley
- *Current:* JWST Cycle 1 GO – “Spectroscopic Confirmation and Characterization of Bright Galaxies at $z \sim 9$ ” (**Finkelstein Admin PI**)

- Awarded \$143,019 to UT Austin to support GO NIRSpec observations of 11 high-redshift galaxy candidates. PI is Micaela Bagley, postdoc in Finkelstein’s group.
 - Person-months for Finkelstein: 0.5 summer months total across FY22-FY23.
 - Start/End Dates: 5/1/2022 – 4/30/2025
 - Location: The University of Texas at Austin
 - PI: Micaela Bagley
- *Current*: CNS Spark Grant – “Solving Reionization with ERMOS on the Giant Magellan Telescope”
 - Awarded \$200,000 to UT Austin to support the development of the ERMOS instrument concept.
 - Person-months for Finkelstein: None
 - Start/End Dates: 3/1/2022 – 2/28/2026
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
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- *Pending*: NSF AAG – “Tracking Reionization from The Earliest Ionizing Sources to the Ending Stages”
 - Proposed for \$628,779 to UT Austin to support investigations into early epoch of reionization galaxies with ground-based telescopes.
 - Person-months for PI: 1.5 summer months over three years.
 - Start/End Dates: 9/1/2023 – 8/31/2026
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
 - *Pending*: FINESST – “Calibrating Lyman Alpha Emission as a Probe of Reionization”
 - Requested \$150,000 to UT Austin to support graduate student Oscar Chavez Ortiz.
 - Person-months for Finkelstein: No funding received
 - Start/End Dates: 9/1/2023 – 8/31/2026
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein
 - *Pending*: FINESST – “The Evolution of Galaxy Rest-Frame UV Colors from $z = 2 - 16$ ”
 - Requested \$150,000 to UT Austin to support graduate student Alexa Morales.
 - Person-months for Finkelstein: No funding received
 - Start/End Dates: 9/1/2023 – 8/31/2026
 - Location: The University of Texas at Austin
 - PI: Steven Finkelstein

Mentoring Plan

FI Oscar Chavez Ortiz will receive mentoring support at an individual level with his advisor, Prof. Steven Finkelstein, and at the departmental level. Finkelstein and the Department of Astronomy at UT Austin place a high priority on the success and well-being of graduate students. We are dedicated to ensuring that all students are welcomed into a supportive and inclusive environment. A number of formal and informal opportunities for scientific and career development will be available to Mr. Chavez Ortiz throughout and beyond the period of this FINESST support.

Mr. Chavez Ortiz's research and career development will be enhanced through a program of structured mentoring activities to provide the skills, knowledge, and experience to prepare Mr. Chavez Ortiz to excel in his chosen career path. To accomplish this goal, Finkelstein and Mr. Chavez Ortiz have developed this mentoring plan which includes career preparation and opportunities to improve skills such as designing scientific experiments, writing telescope proposals, presenting and communicating results, publishing journal articles, participating in peer mentoring, and assisting in community outreach. Specific elements of the mentoring plan include:

Research Advising: The quality of advising that a student receives makes all the difference in whether they are able to thrive in graduate school and achieve post-graduate success. The amount of attention, positive reinforcement, and scientific guidance a student requires varies greatly. Mr. Chavez Ortiz will meet weekly with Prof. Finkelstein, as well as attend weekly group meetings. The one-on-one meetings will focus on identifying daily and weekly goals and provide feedback, while group meetings bring opportunities to present results and engage in deep scientific discussion. Other more rapid opportunities for quick communication include the group Slack channel, stopping by Prof. Finkelstein's typically open door (in post-COVID times), and email.

Nurturing Academic Skills: Writing and presentation skills are essential for success. Students regularly give research presentations and practice job talks at the weekly Graduate Student-Postdoc seminar and weekly research group seminar. A weekly informal lunch discussion of new arXiv papers is organized by the graduate students and attended by all. This provides exposure to the recent literature and serves as a forum to discuss science. Mr. Chavez Ortiz will attend these seminars and meetings to improve his communication skills, practice presentations he will give at conferences, and to stay up to date on the latest studies in his field.

The Astronomy department aims to send all the students to at least one academic conference per year, and funds are earmarked through the topic-based research groups for this purpose, subsidized by their advisors. Prof. Finkelstein is highly supportive of his students traveling to conferences and presenting their own work, and will ensure Mr. Chavez Ortiz has these opportunities and develops these skills. Though Oscar might have limited travel options as a student with DACA status, the plethora of scientific meetings being held within the US is more than sufficient to meet the needs of a successful graduate student's network base, and given the current expansion towards virtual attendance we will work towards remote international participation as well. As Mr. Chavez Ortiz prepares the publications from this project, Prof. Finkelstein will mentor him in academic writing and in the preparation of manuscripts for scientific journals.

Graduate students are encouraged to lead observing proposals as PI at McDonald Observatory's 10-m Hobby Eberly Telescope (HET), the 2.7-m Harlan J. Smith Telescope, and the 2.1-m Struve Telescope. UT Austin has a 67% share of HET time and graduate students regularly lead large programs at that facility. Mr. Chavez Ortiz will be encouraged, when it fits with his research, to submit proposals for observations on these facilities and those with national access, such as Keck

Observatory, the Hubble and Webb Space Telescopes, and ALMA. In addition, UT has access to supercomputing time through the Texas Advanced Computing Center (TACC).

Teaching & Mentorship Training: The Texas Institute for Discovery Education in Science (TIDES) at UT hosts regular workshops open to all educators, including graduate students. The UT Austin Department of Astronomy also participates in the Institute for Scientist & Engineer Educators' Professional Development Program, which offers team training in inquiry-based education using two intensive workshops at a central location and on-site practice. Graduate students in our department regularly participate in these teaching training opportunities to develop pedagogical skills and direct experience in education. The department also runs two summer undergraduate research programs—the Texas Undergraduate Research Experience for Under-represented Students (TAURUS) and an NSF-funded REU program. Graduate students are routinely involved in planning and participating in a seminar series for these visiting students. They also serve as graduate student mentors for visiting undergraduate students. In collaboration with McDonald Observatory, the department carries out a variety of outreach activities including the weekly StarDate radio program and public observing nights. Our monthly Austin Astronomy on Tap (AoT) is organized by students and postdocs and is the largest of any such satellites in the U.S. Mr. Chavez Ortiz will have the opportunity to take part in these activities throughout his time in the graduate program at UT Austin (he recently provided the weekly AoT news segment in Spanish and English).

Department & Community: The UT-Astronomy Department is dedicated to ensuring that all its students are welcomed into a supportive and inclusive environment. For example, the *Association of Women in Astronomy Research and Education (AWARE)* and *Equity & Inclusion* groups meet regularly for discussions; both organizations are co-led by graduate students.

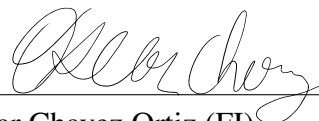
Expectations For Students and Advisors: Astronomy faculty often adopt a laissez faire approach to graduate mentoring. From personal experience, we know that this can leave students feeling confused about research expectations and uncertain about the metrics of success. To address this, the mentoring strategy consists of two main tenets: i) communication - this is key, and we take great effort to maintain open lines about all aspects of being an astronomer, including discussions of career paths, difficulties with classes or projects, as well as general happiness in life, and ii) promotion - networking is key in our field, and Prof. Finkelstein strongly promotes the work of his students, by sending them to frequent conferences to give talks, and by sharing their work with our colleagues around the globe.

Semi-annual Assessment and Individual Development Plan: To ensure student needs and goals are being addressed and that the advisor and student are “on the same page” in terms of their research plans and progress, Mr. Chavez Ortiz will complete an Individual Development Plan (IDP) every 6 months. In this one-page document, students identify their research, academic and writing goals. The PI and FI will meet to discuss the IDP and review both past progress and future objectives. Together we formulate a plan for the next six months to address his goals.

Success of this mentoring plan will be assessed by tracking Mr. Chavez Ortiz's progress through his Individual Development Plan and toward his career goals after earning his Ph.D. in our program. I have read and agree to this mentoring plan.



Steven Finkelstein (PI)



Oscar Chavez Ortiz (FI)

Budget Timeline and Narrative

The budget justification below describes each line item in our request, which we believe is appropriate and necessary for the completion of the goals of this program over the three-year duration of this project. This budget will **begin** on **September 1st, 2023** and **end** on **August 31st, 2026**.

We request grant funding to support one full-time UT graduate student (FI O. Chavez Ortiz) for six semesters and three summers (36 months total) to carry out all of the required work on this project, as detailed in the Project Timeline section of the main proposal. This student will work at a typical appointment of 50% annual FTE effort for all three years as a graduate research assistant (GRA).

Participant Support:

1. **FI Stipend (salaries and wages):** This budget will cover one UT Austin graduate student for three full academic years spanning 12 months each (9 academic months and 3 summer months). FI's (O. Chavez Ortiz) monthly base is \$3,333.33 , totaling **\$40,000** for a **12-month salary**.
2. **FI Allowances (domestic travel):** No funding is requested for travel. FI (O. Chavez Ortiz) will present these scientific results at topical conferences through existing external funding sources.
3. **University Tuition:** We propose to ask that **\$10,000 per year** be allocated yearly on September 1 towards tuition expenses. This allocation of funds will offset tuition costs which is a total of \$19,600.00 per year (this number was based on the 2020-21 Academic calendar year). The remaining annual tuition and fees will be funded through existing external grant sources provided by the Department of Astronomy at UT (i.e. funding from PI).
4. **Other:** N/A. No funding is requested for publication costs. FI (O. Chavez Ortiz) will publish results from this work in peer-reviewed journals through existing external funding sources.

The total budget request for funds allocated to UT Austin across all three years is \$150,000.

	Year 1	Year 2	Year 3
Participant Cost FI Stipend			
\$40,000-12 Months	\$40,000.00	\$40,000.00	\$40,000.00
Tuition			
\$5,000 per semester, 2 semester per year	\$10,000.00	\$10,000.00	\$10,000.00
Other			
N/A	\$0.00	\$0.00	\$0.00
Total	\$50,000.00	\$50,000.00	\$50,000.00